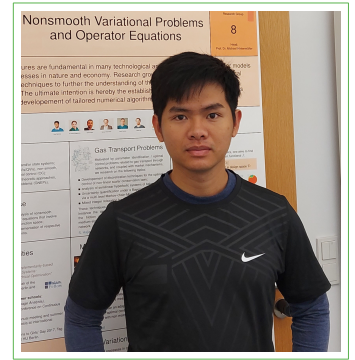


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[Google Scholar/Hong Quan Ba Nguyen](https://scholar.google.com/citations?user=HONGQUANBA)  
[ResearchGate/Hong Quan Ba Nguyen](https://www.researchgate.net/profile/Hong-Quan-Ba-Nguyen)



# Hong Nguyen

## Curriculum Vitae

*"The essence of mathematics lies in its freedom." - Georg Cantor*

*"It is true that a mathematician who is not somewhat of a poet, will never be a perfect mathematician." - Karl Weierstrass*

## Profile

Applied Mathematician specializing in continuous physical systems (PDEs, fluid dynamics) and mathematical optimization. Highly motivated to bridge rigorous numerical analysis with data-driven operator learning to model and control complex nonlinear dynamical systems. Proven research experience in reduced-order modeling (ROMSOC) and independent theoretical research.

## Education

- 2020–2021 **PhD student** enrolled in **Humboldt-Universität zu Berlin**, **Weierstrass Institute for Applied Analysis & Stochastics** (WIAS), Berlin, Germany.  
Within the European Innovative Training Network **ROMSOC** (*Reduced Order Modelling, Simulation & Optimization of Coupled Systems*). Focused on shape optimization, fluid mechanics (Navier-Stokes), and reduced-order modeling for continuous physical systems.
- 2018–2019 **Master de Sciences, Technologie, Santé, Mathématiques et Applications**, **Université de Rennes 1**, Rennes, France, *assez bien*.  
Master 2 in Fundamental Mathematics & Applications within ENS Rennes program (skipped Master 1).
- 2014–2018 **Bachelor of Science in Mathematics**, **University of Science** (HCMUS), **Viet Nam National University, Ho Chi Minh City** (VNU HCM), Vietnam, *very good*.  
Specialized in Mathematical Analysis & Numerical Analysis.
- 2011–2014 **Elementary & Olympiad Mathematics**, **Ben Tre High School for Gifted Students** (CBT), Vietnam, *very good*.

## Master Thesis

- Title *Staggered & well-balanced discretization of shallow-water equations.* [pdf][slide]  
Supervisor Prof. **Nicolas Seguin**  
Description This thesis investigates some staggered well-balanced finite volume schemes for the shallow water equations. A new staggered scheme, called "offset equilibrium", is developed to conquer the *dry-wet transition* issues where almost all the traditional finite volume schemes collapse. Higher-order extensions for this staggered scheme are also mentioned.

## Master Courses

- 2018–2019 **Courses of Master 2 Program in Fundamental Mathematics with Applications, 2018–2019:**
- Sep 2018 Prof. **Nicoletta Tchou**. *Sobolev spaces & elliptic equation.*
  - Sep 2018 Prof. **Florian Méhats**. *Hyperbolic equations.*
  - Nov 2018 Prof. **Nicolas Seguin & Eric Darrigrand**. *Finite Element Method (FEM).*
  - Nov 2018 Prof. **Nicolas Seguin**. *Numerics of transport.*
  - Nov 2018 Prof. **Mohammed Lemou**. *Numerical methods for kinetic transport equations.*
  - Dec 2018 Prof. **Luis Miguel Rodrigues**. *Master 2 Seminar*  
▷ Hong Nguyen. "On the smallest constant for a Gagliardo-Nirenberg functional inequality". [pdf][slide]
  - Mar 2019 **Vincent Duchêne**. *Modeling & mathematical analysis of shallow free-surface fluid flows.*
  - Mar 2019 Prof. **Marc Quincampoix**. *Optimal control & Hamilton-Jacobi equations.*

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## Research Interests

- **Nonlinear Dynamical Systems** ▷ Spectral Submanifolds (SSM), Fluid-Structure Interactions, Model Reduction.
- **Data-Driven Modeling & Control** ▷ Operator Learning, Scientific Machine Learning (SciML), Physics-Informed Neural Networks (PINNs).
- **Mathematical Analysis & Fluid Mechanics** ▷ Navier–Stokes equations, Computational Fluid Dynamics (CFDs), Turbulence modeling.
- **Mathematical Optimization** ▷ Optimal Control, Shape & Topology Optimization.
- **Numerical Analysis** ▷ Finite Difference Methods (FDMs), Finite Element Methods (FEMs), Finite Volume Methods (FVMs).

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## Research & Professional Experience

### Vocational & Research

- Jan 2025–now **Lecturer**, *University of Management & Technology (UMT HCMC)*, Ho Chi Minh City.  
School of Technology (SOT): expertise: Mathematics for Data Science & Artificial Intelligence.
- Jan 2025–now **Mathematical Olympiad trainer**, *UMT HCMC*, Ho Chi Minh City.  
Training undergraduate students for Vietnamese Mathematical Competitions for Undergraduate Students.
- Jan 2025–now **Olympiad in Informatics trainer**, *UMT HCMC*, Ho Chi Minh City.  
Training undergraduate students for Vietnamese Olympiad in Informatics for Undergraduate Students.
- Jan 2022–Dec 2024 **Independent Researcher**, Ben Tre city, Vietnam.  
Conducted independent research focusing on applied mathematics, optimization, and algorithmic reduction. Authored and submitted multiple preprints bridging theoretical mathematics with computational applications.
- Apr 2021–Dec 2021 **Software Development Engineer (SDE)**, *Math+ Tec GmbH*, Vienna, Austria.  
Secondment for technical & scientific training as part of the ROMSOC project. Applied model reduction techniques in an industrial context.
- 2020–2021 **Marie Curie PhD Researcher**, *Weierstrass Institute (WIAS)*, Berlin, Germany.  
Marie Skłodowska-Curie fellow within the ROMSOC network. Investigated optimal shape design and model reduction of coupled systems.
- Mar–Jun 2019 **Master Internship**, *Institut de recherche mathématique de Rennes – IRMAR*, Rennes, France.

### Miscellaneous

- 2015–2017 **Olympiad Trainer**, *Ben Tre High School for Gifted Students (CBT)*, Vietnam.  
Trainer of the selected team of CBT for Vietnamese Mathematical Olympiads (VMOs).

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## Publications

- [math.PDE] Nguyen Anh Dao, Jesús Ildefonso Díaz, Quan Ba Hong Nguyen. “*Pointwise Gradient Estimates in Multi-dimensional Slow Diffusion Equations with a Singular Quenching Term*”. *Advances in Nonlinear Studies* (2020). DOI <https://doi.org/10.1515/ans-2020-2076>. [pdf]

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## Preprints

Under [arXiv Category Taxonomy](#) abbreviations:

### Computer Science

- [cs.IT] Nguyen Quan Ba Hong. “*A Functional Level-Set Slack Identity: Elevating Empirical OOD Baselines into Exact Continuous Combinatorial Equations*”. [submitted to IEEE Transactions on Pattern Analysis and Machine Intelligence (IEEE TPAMI); current status: under review]
- [cs.IT] Nguyen Quan Ba Hong. “*Alexandrov Completions, Simplicial Nerves, and Ahlswede–Zhang Obstructions over Finite Covering Spaces*”. [submitted to Fuzzy Sets and Systems]
- [cs.IT] Nguyen Quan Ba Hong. “*Homological Quantification of the Čech Defect: Quillen Fibrations and Ahlswede–Zhang Obstructions over Finite Posets*”. [submitted to Journal of Combinatorial Theory, Series A (JCTA)]; current status: with editor/seeking expert advice
- [cs.IT] Nguyen Quan Ba Hong. “*Boundary Optimization for Rough Sets: Continuous Integral Decoupling and Intractability Barriers*”. [submitted to European Journal of Operational Research (EJOR)]
- [cs.IT] Nguyen Quan Ba Hong. “*Real-Time Optimization of Streaming Rough Set Boundaries via  $\mathcal{O}(1)$  Deamortized Shadow Registers*”. [submitted to Information Science (IS)]
- [cs.CC] Nguyen Quan Ba Hong. “*A Measure-Theoretic Complexity Dichotomy for Combinatorial Up-Set Summations*”. [submitted to SIAM Journal on Computing (SICOMP)]

[cs.ML] Nguyen Quan Ba Hong, Tran Dan Thu. “*SUAZ-FS: Stochastic Submodular Optimization for Large-Scale Unsupervised Feature Selection*”. [preparing to submit]

### Mathematics

[math.EA] Nguyen Quan Ba Hong. “*Algorithmic Reduction of Elementary Symmetric Inequalities via Invariant Positivstellensatz Certificates*”. [submitted to SIAM Journal on Applied Algebra & Geometry (SIAGA)]

[math.NT] Nguyen Quan Ba Hong. “*An Automated Resolution and Generalization of Rowland’s Binomial Divisibility Conjectures via Algebraic Rank Minimization*”. [submitted to Advances in Mathematics, status: under review]

[math.PDE] Nguyen Quan Ba Hong. “*Universal Gradient Estimates and Exact Free Boundary Profiles for the Singular Porous Medium Equation*”. [submitted to Archive for Rational Mechanics and Analysis (ARMA), status: editor assignment pending]

[math.PDE] Nguyen Quan Ba Hong. “*Spatial Decoupling and Complete Quenching for Singular Porous Medium Equations*”. [submitted to Archive for Rational Mechanics and Analysis (ARMA), status: editor assignment pending]

[math.NA/FVM] Nguyen Quan Ba Hong. “*Convergence and Asymptotic Commutativity of the Adjoint PISO Algorithm on Polyhedral Meshes*”. [submitted to SIAM Journal of Numerical Analysis]

### Pedagogy/Education

[math-edu] Nguyen Quan Ba Hong. “*Beyond Code Generation: The Cognitive Necessity of Analytic Derivation in Mathematical Olympiad Training in the Era of Generative AI*”. [revising to resubmit]

[math-edu] Nguyen Quan Ba Hong. “*Fundamental Asymptotic Limits of Generative AI in Discrete Mathematical Reasoning*”. [revising to resubmit]

### Physics

[stat-phys] Nguyen Quan Ba Hong. “*Topological Phase Transitions and Entropy Bifurcations in Dynamic Equivalence Manifolds*”. [submitted to Chaos, Solitons and Fractals, status: with editor]

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## Books & Other Writings

[bk1] Nguyen Quan Ba Hong. “*Theoretical Competitive Programming (Olympiad in Informatics & ACM-ICPC)*”. [In preparation]

[bk2] Nguyen Quan Ba Hong. “*Programmable Mathematical Olympiads (VMC & High School)*”. [In preparation]

[bk3] Nguyễn Quân Bá Hồng, Nguyễn Quân Trung Nhân. *Some Thoughts on Learning, Teaching, & Research: A Short Novel – Vài Suy Nghĩ Về Việc Học, Việc Dạy, & Nghề Nghiên Cứu: 1 Tiểu Thuyết Ngắn*.

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## Courses

Oct 12–23, 2020 Thematic Einstein Semester (TES): *Energy-based mathematical methods for reactive multiphase flows*.  
▷ TES Student Compact Course: *Variational Methods for Fluids & Solids*.

Apr–Jun 2020 Dr. **Stephan Schmidt**. *Computational Geometry & Shape Optimization*. Master course, Humboldt Universität zu Berlin, 2020.

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## Seminars

Mar 2020–now **Berliner Oberseminar Nichtlineare partielle Differentialgleichungen (Langenbach-Seminar)**, Research Group 1: *Partial Differential Equations*, WIAS Berlin, Germany.

Mar 2020–now **Material Modeling Seminar**, WIAS Berlin, Germany.

Mar 2020–now **Numerical Mathematics Seminars**, Research Group 3: *Numerical Mathematics & Scientific Computing*, WIAS Berlin, Germany.

Mar 2020–now **Modern Methods in Applied Stochastics & Nonparametric Statistics**, Research Group 6: *Stochastic Algorithms & Nonparametric Statistics*, WIAS Berlin, Germany.

Mar 2020–now **Joint Research Seminar on Mathematical Optimization**, Research Group 8: *Non-smooth Variational Problems & Operator Equations*, WIAS Berlin, Germany.

Mar 2021–now **Optimization, Control & Inverse Problems**, Research Group 8: *Non-smooth Variational Problems & Operator Equations*, WIAS Berlin, Germany.

Mar 2020–now **Berlin Mathematics Research Center MATH<sup>+</sup> Events**.

Nov 2020–Apr 2021 ROMSOC Scientific ESRs Seminar.

## Schools

- Jul 26–30, 2021 [Summer Schools on Applied Mathematics and Mechanics 2021 \(SAMM 2021\)](#): *Shape & Topology Optimization*. [GAMM Juniors](#), Graz University of Technology (Technische Universität Graz), Graz, Austria.
- Jul 1–2, 2021 [Durham Days of Analysis & PDE Summer School](#). Durham University, United Kingdom.
- Sep 7–10, 2020 [Model Order Reduction Summer School 2020 \(MORSS2020\)](#). École polytechnique fédérale de Lausanne (EPFL), Lausanne, Switzerland.
- Jul 20–24, 2020 [162nd European Study Group with Industry](#). University of Leeds, United Kingdom.

## Workshops

- Apr 29, 2021 [FUHRI2021: Finite volume methods for Real-world applications](#), WIAS Berlin, Germany.
- Sep 28–29, 2020 ROMSOC Internal Workshop.
- Sep 15, 2020 ROMSOC Ethical Monitoring - 3rd Workshop on Ethics.

## Conferences

- Mar 24–25, 2021 Annual Meeting of the Second Phase of the DFG Priority Programme 1962 ([SPP1962](#)): *Non-smooth & Complementarity-based Distributed Parameter Systems: Simulation & Hierarchical Optimization*. Deutsche Forschungsgemeinschaft (DFG) + WIAS, Germany.
- Sep 14–17, 2020 [Deutsche Mathematiker-Vereinigung](#) (German Mathematical Society) Annual Meeting 2020 ([DMV2020](#)), Technische Universität Chemnitz, Chemnitz, Germany.

## Talks

- Apr 12, 2021 Hồng Quân Bá Nguyễn (WIAS Berlin). “*Optimal Shape Design of Air Ducts in Combustion Engines: Design a General Shape Optimization Framework*”. ROMSOC ESR Seminar. [[abstract](#)]
- Sep 7, 2020 Hong Nguyen (WIAS Berlin). “*A shape optimization problem for stationary Navier-Stokes flows in three-dimensional tubes*”. [Model Order Reduction Summer School 2020 \(MORSS2020\)](#), École polytechnique fédérale de Lausanne (EPFL), Lausanne, Switzerland. [[abstract](#)][[slide](#)]
- Jun 26, 2019 Hong Nguyen. “*Staggered & well-balanced discretization of shallow water equations*”. PhD interview, Department of Mathematics, RWTH Aachen University, Aachen, Germany. [[slide](#)]

## Miscellaneous

- Feb 22, 2021 WIAS-Day2021 ▷ [Hong Quan Ba Nguyen \(Research Group 8\)'s Research Profile](#).
- Nov 30, 2020 [ROMSOC Blog](#) ▷ Hong Nguyen. *Optimal shape design of air ducts* [[pdf](#)].
- Jun 18, 2020 [Prof. Luis Miguel Rodrigues's recognition of Master equivalence & admission to doctoral studies of NGUYEN Quan Ba Hong](#). IRMAR, Rennes, France.
- May 27, 2019 [Prof. Luis Miguel Rodrigues's recommendation letter for NGUYEN Quan Ba Hong](#). IRMAR, Rennes, France.

## Visits

- Apr 2021–Nov, 2021 [MATH+TEC GmbH](#), Vienna, Austria.
- Apr 2021–Nov 2021 [Vienna University of Technology/Technische Universität Wien](#), Vienna, Austria.
- Apr 23, 2021 [University of Vienna/Universität Wien](#), Vienna, Austria.
- Mar 2020–Nov 2021 [Humboldt-Universität zu Berlin](#), Berlin, Germany.
- Jun 25–27, 2019 [Department of Mathematics, Rheinisch-Westfälische Technische Hochschule \(RWTH\) Aachen University](#), Aachen, Germany.
- Sep 2018–Jun 2019 [UFR Mathématiques, Université de Rennes 1](#), Rennes, Bretagne, France.
- Sep 2018–Dec 2018 [Université Rennes 2 - UFR Sciences sociales](#), Rennes, Bretagne, France.
- Mar 2017 [Vietnam Institute for Advanced Studies in Mathematics \(VIASM\)](#), Hanoi, Vietnam.

## Awards & Scholarships

- Mar 2020–Aug 2021 [Marie Skłodowska-Curie Actions \(MSCA\) PhD Fellowship, Horizon 2020 - Innovative Training Network - European Industrial Doctorate Programs](#), Grant Agreement number: 765374 - [Reduced Order Modelling, Simulation & Optimization of Coupled Systems](#) (ROMSOC) [H2020-MSCA-ITN-2017-EID/ROMSOC/ESR11].
- 2018–2019 [Lebesgue Master Scholarship](#) for Master 2 program within ENS Rennes program.
- 2017 Scholarship of [Vietnam Institute for Advanced Studies in Mathematics \(VIASM\)](#).

- 2016 Special Prize  $\Leftarrow$  First Prize in Algebra + First Prize in Analysis, in *Vietnamese Mathematical Olympiad for College Students* (VMC 2016).
- 2015 Second Prize in Algebra, Second Prize in Analysis, in *Vietnamese Mathematical Olympiad for College Students* (VMC 2015).
- 2014 First Prize #1 in Ben Tre city Mathematical Olympiad for High School Students.
- 2013 Second Prize in Ben Tre city Mathematical Olympiad for High School Students.
- 2012 Gold Medal ranked #2, in *Vietnamese Apr 30th Mathematical Olympiad for Central & Southern Vietnam*.

## Computer Skills

- GitHub [GitHub/NQBH](#)  $\triangleright$  [GitHub/ROMSOC project](#).
- Basic R, Fortran, Fireshape, Firedrake, FEniCS, STAR-CCM+.
- Intermediate Antigravity CLI, git, OpenFOAM, LLMs (Gemini's ecosystem).
- Advanced C/C++, UNIX/Linux [openSUSE, (K)Ubuntu],  $\text{\TeX/L\TeX}$ , MATLAB, Python.

## Languages

- Vietnamese Mother tongue
- English Intermediate, focus more on reading & writing than speaking

## Collaborators

In alphabetical order:  $\emptyset$  [filling in]

## Teaching

- Mathematics [Discrete Mathematics](#), [Linear Algebra](#), [Introduction to Mathematical Analysis](#), [Combinatorics & Graph Theory](#) (Summer 2025, Summer 2026).
- Computer Science [Fundamentals of Programming \(C/C++\)](#), [Information Technology Fundamentals II \(Python\)](#), [Machine Learning with Python](#) (Summer 2025, Summer 2026).
- Supervision [Bachelor Thesis Supervision in Mathematics/Computer Science](#) (Spring 2026).

## Students

- In alphabetical order:
- [1] NGUYỄN NGỌC THẠCH (2026) (Software Engineering). *Chronix: Personalized Scheduling with AI and Chronobiology for Optimal Productivity – Chronix: Hệ Thống Lập Lịch Cá Nhân Hoá Tối Ưu Năng Suất Dựa Trên AI và Nhịp Điều Sinh Học*. Current: Software Engineer at Zalo, VNG Corporation, Vietnam.

## Referees

- In chronological order:
- |     |                               |                                     |
|-----|-------------------------------|-------------------------------------|
| [1] | <b>Dr. Đào Mạnh Khang</b>     | <i>Bachelor Teacher</i>             |
| [2] | <b>Dr. Nguyễn Tấn Trung</b>   | <i>Bachelor Teacher</i>             |
| [3] | <b>Dr. Đào Nguyên Anh</b>     | <i>Bachelor Supervisor</i>          |
| [6] | <b>Prof. Dr. Trần Đan Thư</b> | <i>Head of IT Department of UMT</i> |

## Hobbies

- Psychology  $\Psi$
- Literary
- Arts (emotions carriers): movies, animes, mangas, (instrumental) musics, etc.
- Philosophy  $\Phi$
- Writing, blogging

□

[Updated to Aug 17, 2026]